

CTL Key Encapsulation Mechanism x86 Architecture Implementation Self-Assessment Test Report

Algorithm Name: CTL

Algorithm Category: Key Encapsulation

Sub-functions: Key Generation / Encapsulation / Decapsulation

Implementation Version: Reference Implementation

Security Level: 128bit/256bit/512bit

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1. Overview

This report presents self-assessment testing of the reference implementations of the CTL key encapsulation algorithm on x86 architecture. Testing includes functional testing, performance and resource consumption testing, and transmission and storage overhead testing.

Functional testing verifies the completeness of core CTL algorithm components including key generation, encapsulation, and decapsulation, and confirms that the shared secret generated by the encapsulation party matches the shared secret recovered by the decapsulation party. Fixed random seeds or test vectors are used throughout testing to ensure reproducible results.

Performance and resource consumption testing measures the average execution time, average CPU cycles, and throughput of the three core operations (key generation, encapsulation, decapsulation), as well as static memory usage after algorithm loading and peak memory usage during runtime. The number of test iterations shall be no fewer than 100; the actual count is reported per the test command output.

Transmission and storage overhead testing measures the data sizes involved in the CTL key encapsulation algorithm, including public key, private key, ciphertext, and shared secret value.

2. Test Environment

Test Period: June 24 - 25, 2026

2.1. Hardware and Software Configuration

No.	Name	Specification & Identifier	Purpose	Qty.
1	CTL Reference Implementation Source Code	Version: 1.0; Parameter Set: 257-512 / 769-1024 / 3329-2048	Program Under Test	1
2	Test Terminal PC	CPU: Intel® Core™ i9-14900HX; Clock Speed: 2.2 GHz; Memory: 32 GB; OS: Windows 11	Compile and Run Test Programs	1
3	MinGW Compilation Environment	GCC Version: 15.2.0; Target Architecture: x86-64	Compile C Test Programs	1

2.2. Test Tools

No.	Tool Name	Tool Type
1	MinGW / MinGW-w64	C Compiler Toolchain
2	benchmark_selfeval_ctl.c	Performance, Resource, and Overhead Benchmark Program
3	test_kem_correctness_ctl.c	Functional Correctness Test Program

3. Reference Implementation Testing

This chapter presents testing of the reference implementation of the CTL key encapsulation algorithm.

The reference implementation is compiled with the following commands:

```
gcc -std=c99 -O2 -Wall -Wextra -Wpedantic test_kem_correctness_ctl.c <CTL_SOURCE_FILES> -o test_kem_ref.exe
gcc -std=c99 -O2 -Wall -Wextra -Wpedantic benchmark_selfeval_ctl.c <CTL_SOURCE_FILES> -o benchmark_ref.exe
-lpsapi
```

3.1. Functional Testing

Test ID	REF-1
Test Name	CTL-257-512 Reference Implementation Key Encapsulation Functional Correctness Test
Test Objective	Verify that the CTL-257-512 reference implementation supports key generation, encapsulation, and decapsulation, and verify whether the original shared key is consistent with the decrypted shared key.
Prerequisites	1.CTL reference implementation source code is deployed on the test terminal PC; 2.test_kem_correctness_ctl.c can successfully invoke CTL key generation, encapsulation, and decapsulation interfaces; 3.The test program compiles and runs without errors.
Test Steps	1.Navigate to the CTL-257-512 reference implementation source code directory; 2.Run make all to generate test_kem_correctness_ctl.exe; 3.Execute test_kem_correctness_ctl.exe; 4.Verify that the encapsulated shared key and decapsulated shared key are identical; 5. The algorithm will be executed three times in a row and generate the ctl_kem_kat_vectors.txt result file.
Expected Results	1.test_kem_correctness_ctl.exe runs successfully; 2.Terminal displays "All CTL KEM correctness tests passed."; 3.Test vector file ctl_kem_kat_vectors.txt is generated correctly; 4.ss_enc and ss_dec values in the test vector file are equal.
Actual Results	The program runs successfully and displays "All CTL KEM correctness tests passed." All three groups of ss_enc and ss_dec in the ctl_kem_kat_vectors.txt file are equal.

algorithm = CTL-257-512

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

99B4BD408EE8A95011ACEA41FEE592AF5879CD3D3EAD420122FB50AC7B94D5F15091E56DA6D
57C7A10E2766DF8144E94A2D67DE507A6139B0DA5DC0B72A106A1FA49413C8612D88938AC9316
DC8A0EFB45464AD04FDA893F76B94333FDA9976DFBFD39CDC7CDB2BC071FD0F98D5551D5D35
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36615FA02CD2082B0C33D7CA23014D918331530BE5E0D00598F7AF0B93E55A9151146B3D80889682
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sk =

19AE5D933F84538C0270359F7DED4695A38B68A9E050084EFBA8BDCDD685B942EE3FE05922EE90
1FE0E9E956242B81B7F61C15C1111D711EAEB4076508CEC087DD0000F011FF00F0011010000F00001
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F00F000000001001000100110F1000F00FF0F1F0F00F0010000010FF00FF000F0000F00F001F0000FF001
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7E97DC02A7F1D94B3C3D54B994B59D2F48C30B6DDCFF0253E781E573CD09BFC1D20076C4587A4
F85009894AD762C19D7EE5FD0356B546F654705DEAB6426B0916A05057E176F2A00C5B2D34900920
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ct =

D98D3DAAFB6D25886B47F191AA1E96323CB0B48B185E1B7F0DAB551ADE7C6A0B7DF36741428C
BC7DE0FC30FEF88B3F62F4C11C82E28A7BA66E94BC4B099E72015F2CE8A95EC2A54371FA6535F1
696A7D385D9373B2705133E05CE35E3D8D350E5B06CC20CC451368F31D1C6E8D29B7F63C7F6CCA
2A19397E594B6FC51388B3A575C29A25BEEE271ADCE66477ABC252B7DDB9ADC3151663D8499EB
3FE32FBBE8EBB722E47DC16F542D139B939B635C1A77B9245E84EB4FA7DE57494A9EF6674707763
4CBFE531C6D231C23A292E39B4A3BCF5DDEAF27463F2AA1BAC9B643A840E93FCC54F4323F33E
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8F319CC8311745E685099A7CEC230B73F937E496B2D60B39F7FA2B419678B70A02114A91843DE800
F7AA58F6FFFA992AB3F847CC0A1E4B54DBCA405E65BBF0D577D01AD73F84D3E801A99157AC2A
3A3AE1E78A39284A9A0736242014B56AA4147D930F8C2826D080B9CE114A873906B90B1AE9E9876
E054B8D0C284

ss_enc = 460FE91C8D752E7BA74838C7B2B891DD

ss_dec = 460FE91C8D752E7BA74838C7B2B891DD

result = PASS

case = 2

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

991581121AADB24C53C4E3C3ED26E30087DDA5E64C6C24CE94963F7C4BE3EBEA5C89F63B141F3
B9B1D37FB499539AD64C53B5502F1680A15631F6C98F730431373C9A54EFBA63DAB090E9725C196
1EE1CD771F4FB93A9E5786D8F1FA74BF773C4C4E48384E6CAE5392417570B9B85E49E1800BC4469
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B7FA5590D86C40CF4F9A8FC5D97E7BEAF4898385C05FE44C0560D7137DADEAE8BEA2A003F406
AE1B7E66F847080A86F56ECE1142826DBCC33D392F46495CB9B6DA18C5A16E38DE256C3CA1F20
AD186A4207A91B5EC6BBC546C963B9D645CA14C5A8255935277F87FC4993F6AEACCEE1550915C
4328A34F2D50E6A758035CC20718A915A679B31460BA273B3F64B19DBC8372A0999949AD412C290

E3205752F8AC15AAECF02D06F776FDD508418B8DA406229E53301F1170D1D026F8562B13E3B6C10
CDDBC2266F66AF635742D3E3568ABF55F0B437B732B81FE89A9D4A126DEC5EA5CEEB266EF9DA
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sk =

19C10DA35D778BACBD25A9A47315A4F312EFD4EC044C315DC4CA00F59E297A9F8D1A79ACA146
51B4FC7482B26D2F7125FCBEED20025EE7947A5AE5FE04F9B6681EFFFF000F0F0000000001F1FF1
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3C08403FF8307DF7E0010BDDFFFC0FFD0BD04103BE8207D0800C113D1BF046FFD17FF7DFBC0C0F
8013E002F7ADAF5EFFCF43C9857EE67782577FA9C56DDF887EB8E81E2A1A782E8A877F37334013
DC195758408CA5EEDD3CAA1234F96D60351C7B58EC1472519B6125588EE448E89D1B3ECD369B3C
20B59BA1CBF7929804ABA0821CA3474168C0641118DEA40981458704F3AF5E69237C4E21AAE7F92
F18EBF8517D48F15896878F985DEA161081E88770BD644A72588732A7984021EA06721819C95DA2F
672FF9F495C497E0C19EF59648C3BCA269F5FC19FB7B077CBF484E6F39163892D44D9AFF60991B05
FA582102CB86B198A345E5D7340579D770CE5DFFCD8126B013E13A29C5D41E814703121B6240408
09F0707FA297AC2DF76134319E5DACC5F789C23A56225441A0D1843F7A9A0589FCB064BC8A117F
CAC3223F30882EB1F101FCF50B96B49A86D48D1926FA9BE7DDCFB405D6F63A088304436D16F65C
0B9AE1FAA0345F9F2B85B3E30550D2C96203D0505D7D31347CD1B12C46F40D11A11F1CC511DD2

C1E4F6FD30980E160B09810FDD71F14764205CA1E08004339473EFECE2D5C7CDFF731088BF327FF
0C9A5D6E9D0111208A687DF31DDB88830B44C0A50D2EBE4F905FE7C14621C8925DB8C4C07FE373
E1433838CC27DC222503AB81C87BF0D0D05E3DECE5161F1FCA357F39F0477AE3F3A177F93009C4
4E17D254F5AD313F3BD1F3186FB37CE8AE035CDE817F0775BC1988CD47B6D507EFD2522FF66AF7
EEF40C65DA030E71316F5397A346EB9115C581776494CC3A58A1F72CF22925258D741B383B3AD77
E715E92E876E3B45DD2F47A4009BA651A2264C7A92FCC8A39183E9867D5FF957ECD3290A1B6F34
1E852500A701A59835E820F0FAC093E909C39A7CD10BE9337C8E7BED451E3D0A16B15E39BD4251
978378A6DC0AAAE5039F333162E551817631BE34571785E025B28CD877BC054BD0D71797088D0FB
092F51C7BFB7735189DA07BE82F47BB5A5B1408EAEF0AFB2F99542D29FF284CB9DFBFD3035F43
B7CBA19BBDD9D971540366BCA762B2BF9FBFB6EFE2E0051BE8A27C2FFB0FFD4D78B2DC20A45
E83DCFF72F94AB8816C2312999CE85FA20269C7E764133591FB576084A779E3EC7DDBB4F442190C
C54FC4BDFE4D776800087FD309321BE7C2C9BF9842AD23C8BEB8630D9FAF232881CF49D50736B1
81C7FBC4D8741AAAC4417181F57E79F8F196F17DF4A3745E2204DD917D0E46A02D52D1E123AFD5
8DB6A62CA21534FEFD4D244CE5D94F240AFF90C9C283829C46C05FAE8E3BF8E5A79E6846282143
1EE68024E5EDAEE2376D6CCA96283F024B15B47CE59FE9526F763EF35B3A49CBEF45EBB7F5AFE
21C51ED26759F832BAC4E1A7BFD7B79ED9E5743DFCE472F3BB3E5F0392279D1FDA83E7A3256CF
9A1BFB22063DF5F09DD54FA622E3F3B9468488F09479372F5690894C59D96ED03EDED120185C2337
297E8B5142FB9D1762AEB67783906E5866C6B9831B82A1DDBB73DE4FB3405F4DE739D8CB697688
2285BAE0A3CE09842FD0332605E90AC8020620671FCD6EBA4B8BB42635F015E88D81994486615811
21AADB24C53C4E3C3ED26E30087DDA5E64C6C24CE94963F7C4BE3EBEA5C89F63B141F3B9B1D3
7FB499539AD64C53B5502F1680A15631F6C98F730431373C9A54EFBA63DAB090E9725C1961EE1CD
771F4FB93A9E5786D8F1FA74BF773C4C4E48384E6CAE5392417570B9B85E49E1800BC44697A73EE
F59C550B34D8509FD3E3B878A74A3BEA2E681C900DB223F2BDE9CF48A56A945729486E7EA53567
320E28719AB60AB6DC218C85822BD0E5814AB3828656A180B93F22B6BEE3851234EE914407E4B59
A6E7D59A604C8CACD5233B5EE7D5DEBCAF0FB8949C917DA888C0A366767058E64022C0F6DBE3
6AA3DE5E0EC9C9891C11E86319FB71090EF54E588B3A30C14626FAB7888FD04440C94A178B7FA5
590D86C40CF4F9A8FC5D97E7BEAF4898385C05FE44C0560D7137DADEAE8BEA2A003F406AE1B7
E66F847080A86F56ECE1142826DBCC33D392F46495CB9B6DA18C5A16E38DE256C3CA1F20AD186
A4207A91B5EC6BBC546C963B9D645CA14C5A8255935277F87FC4993F6AEACCEE1550915C4328A3
4F2D50E6A758035CC20718A915A679B31460BA273B3F64B19DBC8372A0999949AD412C290E32057
52F8AC15AAECF02D06F776FDD508418B8DA406229E53301F1170D1D026F8562B13E3B6C10CDDDB
C2266F66AF635742D3E3568ABF55F0B437B732B81FE89A9D4A126DEC5EA5CEEB266EF9DA60851
CCBD3E522E3429600E8D552AD2

ct =

D95C4D11C8C700528D3B2F5F2F30BA1ED3D4476B23F7B92E711B412B5C6BB72EC5FE7578B9B891
236EAE25F198AD9793241FF28DEA7C2A0DE09537CC4C23B82E5E6E98818970CD76606F8E673347
1F26A476AD257A8C7F5D54A733F1EA399A25AAEDA311889C11E23C200418DD74EFAD09624A98B
15C678CC993378009150D5981D6E1F484BF3A415B35B39E5EB6A3291DD176A4B95494D50E71656A
3E03C3E60381A16A345C5E5A2C0658011ED55C698C4B2884B37B2388160D2CFD94D3345547A61D
A6CB362094DCDF00B4A1A6A8BF3ACECF8D9579AE56BF913E41570A9A585C7737B42E383D225C
B7D4203E724C980677626D35960AD4E77BC6410FE26125D54B86AFC262CAEE30C05E318978AE2A
44481B36874A523FE49982F5AC86DEBB091F88DB27D348B2E24980E6C399991D4DB683AAACDEE
B44FF53110514E0C411203821BCA3AAD4203BC7E34FA9921F9FB5BB579F3291084ECC17EE96B5EF
5D0001294DC01EE082E985488602A77E6CE77D6398A0610954D766206FF25E75BDB502CAA6C027C
84AD5A06317A13EC69AF9D92534A2CB42D06C65B70726876EB50FC41322FFBDBBCA706004B2D8

DA89DB7E9D1D7431B97918445CFB3194DE3B9AB6E22657CEFA61B3702446613FC2E0600FA319C9
6CB3DBAC268

ss_enc = C5FD9F8003010BDA9A156C81FFD8B31D

ss_dec = C5FD9F8003010BDA9A156C81FFD8B31D

result = PASS

case = 3

pk_len = 521

sk_len = 2953

ct_len = 473

ss_len = 16

pk =

998B97C194C0DD837E7803595B40DDC4A6E11507EEF4270CBE8C74BD12509A80E06A0D49D50B6
B83A05093BB09D63F2D1B1FE7E45FC0AF742D03BA43D094C18B1A60E65C5C3698E5C735BD74905
F6BC51E078A16D9C3A45D42818B587777FCCCF083A0236BD9E6627D33BDE2AA0AADE4B16C42B
AF89FEF0A7ADBFF99C8269DF716D77328CF0C5BF54CE6A69F9762D763C40DBE52E0C015EBDA70F
597F50D71AFDFC7BF0BA48237EAB8995CBE98FA201220A7CD7A7E1E652C3257C1C015BB52FDC
B1907304337AC0B78F087DBCEAB3972F0E8FAAB5D2678410C9E7F91FCC38D60F245C3C55BA9C0
4A0B6DC42C6D251AEE4839E22788C68173BB3AC39058DE6133368F3CB30C08AD70C97D14DED9E
845F591EDC9E5342EB19CCCB628E46FD7C58D259CAB627B2C3A021731D2C9C2572BBDA40C55F
A4712303EB94524AE92984ABB93E3D813F2B3049ECCBAF59B1473427A2C9953A0C52657A411E672
B97FE45C7EF11DF968A77295EBEF47075BDD336836DE4739F194B782091B227EEA110821ED87C73
1E0294AE18E7DCC620662A0515BF41F5D9B7EFA31BC4527364DF0BEA2B6FCDA41E66913C585819
861A16AA9DFB0FD5BFE9828E98FD6DBFA4CA5BA7CA521715B9DFC0BC37CF0C43DFC44CFC1C
04F333B75398C17893147017A80066BB96FF3BF454A40C888588FE873AE8570A11FF7071A70992C00
7036414C094B0DF61E6044E1447668A8074A2

sk =

19DB89239991D32C3EB3108A23D1421031A8D9487C3780776C30D933302760C3740655D84B01E4B3
1CE8F5FDEE479BF63F572106F3D0DC0062E114C2569F6225BA001001F000F10100F0011010F0F1000
F0F001000F0010FF0F0F001F0F0F01F0000F00F0000101F0F0E000000000000F11F01000F01000010001
1FF0010000F000000100000F00010011110000101000000000F00000111000F10FF000000F000000000F0
00F00000001000100F0F11000FFFF10FF00000F00F0001001000F00000000000F0100F0F00F11010000F1
0000000F000000FF000100001011101F1F10F0F00F1001000000000F0F100F10FF00100010001F1010F00
1010010101FF010F1000000011F00F000000000000F0000100F0F00F0F0001F000101000F0110000000100
0001000101000001010F1010010F1F011101001F0000110F000011F000F0100F0000100F000000101F010F
00F000F0F0000000000F010F001F000000000F0F00FF0100010100F00F10F000F00000100000000100000
00000011001FF00100F0100000010F10F00F010FF01F1000F0100F000F0F00000F00FF02010010FF11110
01F1000F0F000000000000000000F0011001F001F000010000110F00000101F10000110000101F01100000F

E0B1742D45D0B1F0A96E0782AA4780AEFF0ECCEB70F188910AE595E9B8EBC6E3B0B5108D6B9D2
6070ED24D077B8B324681017D6EA6575E0E29B2C14F0C6246E543D8C1DB41346F51BE003E2A9D78
CA4158F98510E761034F1E7FDEF831976AF308CF7FE15D1749A3DF571829F413B29569F58F7B27EE
0A81CEA74A47D67430B211D1FA139E36EF55669CA056C864DFD110876533474643F8B97C194C0DD
837E7803595B40DDC4A6E11507EEF4270CBE8C74BD12509A80E06A0D49D50B6B83A05093BB09D6
3F2D1B1FE7E45FC0AF742D03BA43D094C18B1A60E65C5C3698E5C735BD74905F6BC51E078A16D9
C3A45D42818B58777FCCCF083A0236BD9E6627D33BDE2AA0AADE4B16C42BAF89EF0A7ADBFF
99C8269DF716D77328CF0C5BF54CE6A69F9762D763C40DBE52E0C015EBDA70F597F50D71AFDFC
7BF0BA48237EAB8995CBE98FA201220A7CD7A7E1E652C3257C1C015BB52FDCB1907304337AC0B
78F087DBCEAB3972F0E8FAAB5D2678410C9E7F91FCC38D60F245C3C55BA9C04A0B6DC42C6D25
1AEE4839E22788C68173BB3AC39058DE6133368F3CB30C08AD70C97D14DED9E845F591EDC9E534
2EB19CCCB628E46FD7C58D259CAB627B2C3A021731D2C9C2572BBDA40C55FA4712303EB94524A
E92984ABB93E3D813F2B3049ECCBAF59B1473427A2C9953A0C52657A411E672B97FE45C7EF11DF
968A77295EBEF47075BDD336836DE4739F194B782091B227EEA110821ED87C731E0294AE18E7DCC
620662A0515BF41F5D9B7EFA31BC4527364DF0BEA2B6FCDA41E66913C585819861A16AA9DFB0F
D5BFE9828E98FD6DBFA4CA5BA7CA521715B9DFC0BC37CF0C43DFC44CFC1C04F333B75398C178
93147017A80066BB96FF3BF454A40C888588FE873AE8570A11FF7071A70992C007036414C094B0DF6
1E6044E1447668A8074A2

ct =

D93290FD94A2C273DB3BD306FEB89A4D160733DF7E87361E96248F182A67CDC938FE415CBE4340
31D288090BDFD18F12F341862F0FD3C4F19AD8880555BD1A750D46D6FEE6FA81846BE3095768FE4
11A42F53A19FBEC2E5E38466678E43A20C2A5A3540007D34FB273AF66CA99F6AB26528C624DC85
E590AB276ECE6774CA65D9FCF8B196ECCD8454BF3D771999176FDB7659ECC5180A2D8CD9569B7
8A626E331B1046817D639D17C7D6CF45D02271ED0999DCB775BCEBE53D0EB4943271876A8904EA
67CD03FBFFFEFFCCC9C7C076BD42B150274D131D2722DD668B33934153C6248AAFC12D50A3F2F0
6FE104B2BFB2E3B171C7109FE7245A34E73D301BC1F6A07E129A109FD182681A5780E7C8A3AB6D
039ECFE8753AF3FC83EE0C6D52E79BD47B6F1522384B42E49ADA156F2EBC6762CB3B32201C79E4
9470D978268353A553162CA343B3FD01BF33E1EBC7141DCA9F19E2DD103220921CC8EBD905E58C
804C9F632DD3560BF78A21513DE403BDEDF4AA95EFE492C96AB37B739263ACE66BF0383915A036
D9F97E91D1F285C87688B0B1E793B8EC478517DFE074B3D456896C53392A846259B88C45BF71AF8
1200DA5853B796F9117AD03450DF117073203B1EBD19F7B67207808F6F71CCA8ABC211921C98F8F
96C6D1B29

ss_enc = 28A57455611066AE6E52E5E7A2604445

ss_dec = 28A57455611066AE6E52E5E7A2604445

result = PASS

Test ID	REF-2
Test Name	CTL-769-1024 Reference Implementation Key Encapsulation Functional Correctness Test
Test Objective	Verify that the CTL-769-1024 reference implementation supports key generation, encapsulation, and decapsulation, and verify whether the original shared key is consistent with the decrypted shared key.
Prerequisites	1.CTL reference implementation source code is deployed on the test terminal PC; 2.test_kem_correctness_ctl.c can successfully invoke CTL key generation, encapsulation, and decapsulation interfaces;

	3.The test program compiles and runs without errors.
Test Steps	1.Navigate to the CTL-769-1024 reference implementation source code directory; 2.Run make all to generate test_kem_correctness_ctl.exe; 3.Execute test_kem_correctness_ctl.exe; 4.Verify that the encapsulated shared key and decapsulated shared key are identical; 5. The algorithm will be executed three times in a row The algorithm will be executed three times in a row and generate the ctl_kem_kat_vectors.txt result file.
Expected Results	1.test_kem_correctness_ctl.exe runs successfully; 2.Terminal displays "All CTL KEM correctness tests passed."; 3.Test vector file ctl_kem_kat_vectors.txt is generated correctly; 4.ss_enc and ss_dec values in the test vector file are equal.
Actual Results	The program runs successfully and displays "All CTL KEM correctness tests passed." All three groups of ss_enc and ss_dec in the ctl_kem_kat_vectors.txt file are equal.

algorithm = CTL-769-1024

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

AAEDB9E409B79C235DEBA6776DAB251B5E7B60A973DDFAEC4C60ACC7529CA05E43356A1FB9
B066A2C9773E8E084FE2FCC05326C2C93DAA35B0A4145BC72AC251CB8BD997ED518CAEF173811
FED4E5BF320AF90670B5F8FDD7949E4C62EFF971F575B76D80582E3E29352DBE23AE1AA85205C1
6C849DBAF150FB8E831E018A503C24781F00F3EBF9D34F32C38DD71572794D152102F5AB2DFD3E
F74EBCB4511151C61A6ABE00263AE7EC0B717CF11D5F56F1A6F291E176ED9DB1E4653ECFF719F8
B8EEBE16419EA4DF190604FF5EBAFE4D056CEA3B82D1F82B915581559175E0658682906A377A30D
B0CD67F1708CC7C62C00BD9E8754E7A2433A94EDC2F94546916ECEE5916E822FB478E941046C2F0
5B8D09FA1AFF6343B0FA9D472B28FF30383A70ADEBDC9DB655AABE051A81B7679AA875509571
88A29AAC3C81F677295BDBA9CB3CCAF16F8D5A6BF2EA3046B18F149A4FF7F778A0D3BD458C0E
ADAE4E5F89699BA0807727FDD6685A428FD128F5C9A9134FE155B0C198F31C9071410C95BDD1A2
2495894E1AFA8C7294A0FFA20A0385343812EFA856FF5BF1E0687435A0BD621FCF6857CB8E44388

05486C2D07CC650DC876A91C31159D72E3152D3D25CA5D141A2E6D3B48435BA64738C0E606E883
9E89B36DB62B98D8A5964CEA454C9AADBA2E31B32EF359AD24C3562516B6DBF869E0D12AC947
E7130C268B49B9529BF23FF1F6E3D0D7CD17E843E2C9F30A0183007A2ED8CB1B881A246C8A8AC5
7F143797E8C95503CEA0EAF1DEA02BA8524149206BC4812D0D55BFFD8033E934FA764B6CFBB428
885133F9C7911DE1A7A8F6C5479689BD348116115FC7611DF8ABC70B0FF5409D41435A5783E0F292
177B55BAB65576997165C50636929CC24000EA99C99A619B4104F1D384CC839A834FD3CCABDAF3
46A79C467D41019F100FD766B6AADC4E25953FBB766B69378EF6C5C57B511C7B67C3BF61D67318
79DFAC50C108109CBBD0D4364833DC6AA422F100CFC195FDF9DFDBEBEDD297F3D0F9783B7AE
1F23BE3D09BC53AD468B52EB86517E864AB50CF61E58D3CB069D6E74FC397BEAAFFD818A7D3A
CDB34EC54CB051C40A18217190E298D5A3A5C2F6EE9FC666106C7392B21E96310D9B674590A4110
955727AD6EC77E2A4EEDFD050F2ABB4C3CD23E1AB84663096D85883E5E9AFCDA797DDDE255E2
5883C76B78C6453AA7979280F3813E4416118F6717CB33D5C27FB6E265FF1461A1C18EC30F3357D1
E54417A0CB5044F1DD7F25E0A9B472E6FCE08E3824DE1BF9794655A61F7374303DE6597DE207311
019328452D0D0F07635FA51B73D5FFC6ADF3CBE785B5CB6939E88896CD2C331C7FB3240789B6289
3DECAA564F72F85CF26AC69AB582F43470CC0A3C0BEED8B563A779D742465924D8D807C83928A
AD35CB06BB88D3401A74E87C0441B47FA3C9560E17D6C52BD3A37CF48C9FEB80AFE263B2E6849
BB4B820DF1A3AB8934BE17BF52F110CA57C7A6A9AD6DF8594F51608164BFD0601B07870B864FE8
B1B088D11031E51910D1E705027AEAF1FBEE109773709A3EE7298A0319DCB7720870C3C03755FFE
9A332F0D73943078EFD56367A98702B582BBDFD7A4BC44649C9C8CFC584E3A9A29E95E24A73445
75DD2421CA8834BA68027BAD42D2C26AB847E68CDC1B7E6D5EAB2D5501044685CFBC9D90948E
A3199FC10902617964B454F0BFA290CE439721374A59556BB01B1

sk =

2AE87E15F7E6CAB66514C99994C2F5BD92BC52A8554D2C081C0FDE814A57C2964B88F2D26D1A7
6C7C4E199431B7C311A7BEDD9E26E7BACEE31120166F61BB2756B1110FEF01101001011FF10001F0
1F1F0F00FF0F001000000F10FF1F01001F0001F1100F000010011F0100F0010F0100100000110FFF00000
020F00FF0010201F000101F100F00100010000FF1F00011FF10F1FF01F0011100F0100F0001F010001001
001012F0F00F0F00010000F000000F0102001100012F10100100F010FF01FF001F00F001F0001010F0110
101100F0010001110F0000101F1FF000001F01100F11F01100001F01F10F011F10000F0E01000FFF10000
10F000F001011100000011F0FF011000FF000001000F00101F001000F000FF0001100010F010F10F1F000
1010F1100000000010100F11110F0011F011101210010F0FF00F0100000F00000F100000F1F11F000000
00000000F0100000F00110FF0000F0F11F1110F0000FF00F0010F0FF000F010FE0010100000100F110000
1000001000F0FF10F00100F0000001F11F0100010F0000000F00F00F1F0F01F01FFF0100FF11000000000
001F0E010F1000000F0000000F100101F000011F01F0F0F0111F20001000F00F0100101F0011000001010
0100000010100001110F1F010000F0001000000FF1010011010100010F01F000000010F00F001101001000
000010001F00F20F0010FF000F00F0FF0100F10101F200F001F0100F00010000101001F00100001000000
F1000F101F0FF00F00F1000010000EF10000010000100000F00000F0100F01000F00000F00F100010100F
0011F01F110000F0001010000010F000010F000F0111000F010F0E101000000001F0FF0100FF000F002F0
00F001F10011100F01011F00E00F10011F0F00F1000FF0F0000000100010001E10010101100F001000010
0021201FF0F0010F0F00F0F000101F1100F001100FFFFF100F01001FF0FFF01000F000000010111000000
F0011010000100010101001010000F00001001F10010F0100001F000F1100000100001F00F000010FF0011
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result = PASS

case = 2

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sk_len = 6030

ct_len = 1006

ss_len = 32

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ss_dec = BB7E10C333862EFFB7C4C78F68CF8A2CB878D9A5017B00ECBF8A4F4FD5B72ED7

result = PASS

case = 3

pk_len = 1230

sk_len = 6030

ct_len = 1006

ss_len = 32

pk =

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ct =

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68905DBB91389A0D3BB2B333D14D9CFC3A2041B45B4CE500BED723F20E2

ss_enc = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDDAAC

ss_dec = B11EF9E97DDC18BE412371D4279CB7626625E9B8459D5E25B0D625C289DDDAAC

result = PASS

Test ID	REF-3
Test Name	CTL-3329-2048 Reference Implementation Key Encapsulation Functional Correctness Test
Test Objective	Verify that the CTL-3329-2048 reference implementation supports key generation, encapsulation, and decapsulation, and verify whether the original shared key is consistent with the decrypted shared key.
Prerequisites	1.CTL reference implementation source code is deployed on the test terminal PC; 2.test_kem_correctness_ctl.c can successfully invoke CTL key generation, encapsulation, and decapsulation interfaces; 3.The test program compiles and runs without errors.
Test Steps	1.Navigate to the CTL-3329-2048 reference implementation source code directory; 2.Run make all to generate test_kem_correctness_ctl.exe; 3.Execute test_kem_correctness_ctl.exe; 4.Verify that the encapsulated shared key and decapsulated shared key are identical; 5. The algorithm will be executed three times in a row and generate the ctl_kem_kat_vectors.txt result file.
Expected Results	1.test_kem_correctness_ctl.exe runs successfully; 2.Terminal displays "All CTL KEM correctness tests passed.";

	3.Test vector file ctl_kem_kat_vectors.txt is generated correctly; 4.ss_enc and ss_dec values in the test vector file are equal.
Actual Results	The program runs successfully and displays "All CTL KEM correctness tests passed." All three groups of ss_enc and ss_dec in the ctl_kem_kat_vectors.txt file are equal.

algorithm = CTL-3329-2048

seed_len = 55

initial_seed =

ADBE8B8491E2FFC8C5D6233C091A17607D4E5B54A1B28F9895E6F3CCD92A27300D1E6B6471425F
A8A5B6839CE9FAF7C0DD2E3B3401126F

case = 1

pk_len = 3009

sk_len = 15617

ct_len = 2353

ss_len = 48

pk =

BB2A133D8802A109CE74A90B6C0848479563E4BD364A0A5D6C720C77388B1144BF3F4A6AED8686
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ss_enc =

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ss_dec =

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result = PASS

3.2. Performance and Resource Consumption Testing

Test ID	REF-4
Test Name	CTL-257-512 Reference Implementation Performance and Resource Consumption Test
Test Objective	Measure average CPU cycles, throughput, static memory usage, and peak memory usage for the three core operations (key generation, encapsulation, decapsulation) of the CTL-257-512 reference implementation
Prerequisites	1.CTL reference implementation source code runs correctly; 2.benchmark_selfeval_ctl.c can successfully invoke CTL algorithm interfaces; 3.The test program compiles and runs without errors.
Test Steps	1.Navigate to the CTL-257-512 reference implementation source code directory; 2.Run make all to generate benchmark_selfeval_ctl.exe; 3.Execute benchmark_selfeval_ctl.exe to measure total time, average time, average CPU cycles, and throughput for key generation, encapsulation, and decapsulation; 4.The program records static/baseline memory usage before algorithm execution and peak memory usage during multiple runs, then generates ctl_kem_benchmark_raw.csv and ctl_kem_benchmark_summary.csv result files.
Expected Results	The program runs successfully and outputs algorithm performance test results, including average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations.
Actual Results	The program runs successfully and outputs performance test results. The program measures average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations over 100 test iterations. Key generation: 7,694,794 cycles/op average, 314 ops/s throughput; Encapsulation: 29,607 cycles/op average, 81,973 ops/s throughput; Decapsulation: 159,884 cycles/op average, 15,138 ops/s throughput.

	Static/baseline working set memory average: 5,712,865 Bytes; Peak memory average: 5,717,524.48 Bytes.
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Test ID	REF-5
Test Name	CTL-769-1024 Reference Implementation Performance and Resource Consumption Test
Test Objective	Measure average CPU cycles, throughput, static memory usage, and peak memory usage for the three core operations (key generation, encapsulation, decapsulation) of the CTL-769-1024 reference implementation
Prerequisites	1.CTL reference implementation source code runs correctly; 2.benchmark_selfeval_ctl.c can successfully invoke CTL algorithm interfaces; 3.The test program compiles and runs without errors.
Test Steps	1.Navigate to the CTL-769-1024 reference implementation source code directory; 2.Run make all to generate benchmark_selfeval_ctl.exe; 3.Execute benchmark_selfeval_ctl.exe to measure total time, average time, average CPU cycles, and throughput for key generation, encapsulation, and decapsulation; 4.The program records static/baseline memory usage before algorithm execution and peak memory usage during multiple runs, then generates ctl_kem_benchmark_raw.csv and ctl_kem_benchmark_summary.csv result files.
Expected Results	The program runs successfully and outputs algorithm performance test results, including average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations.
Actual Results	The program runs successfully and outputs performance test results. The program measures average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations over 100 test iterations. Key generation: 30,364,861 cycles/op average, 79 ops/s throughput; Encapsulation: 62,111 cycles/op average, 39,012 ops/s throughput; Decapsulation: 316,103 cycles/op average, 7,655 ops/s throughput. Static/baseline working set memory average: 5,911,744 Bytes; Peak memory average: 5,922,570.24 Bytes.

Test ID	REF-6
Test Name	CTL-3329-2048 Reference Implementation Performance and Resource Consumption Test
Test Objective	Measure average CPU cycles, throughput, static memory usage, and peak memory usage for the three core operations (key generation, encapsulation, decapsulation) of the CTL-3329-2048 reference implementation.
Prerequisites	1.CTLReference Implementation Source Code runs correctly; 2.benchmark_selfeval_ctl.can successfully invoke CTL algorithm interfaces; 3.The test program compiles and runs without errors.
Test Steps	1.Navigate to the CTL-3329-2048 reference implementation source code directory; 2.Run make all to generate benchmark_selfeval_ctl.exe; 3.Execute benchmark_selfeval_ctl.exe to measure total time, average time, average CPU cycles, and throughput for key generation, encapsulation, and

	decapsulation; 4.The program records static/baseline memory usage before algorithm execution and peak memory usage during multiple runs, then generates ctl_kem_benchmark_raw.csv and ctl_kem_benchmark_summary.csv result files.
Expected Results	The program runs successfully and outputs algorithm performance test results, including average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations.
Actual Results	The program runs successfully and outputs performance test results. The program measures average CPU cycles, throughput, static/baseline memory usage, and peak memory usage for key generation, encapsulation, and decapsulation operations over 100 test iterations. Key generation: 1,585,861,935 cycles/op average, 1.5 ops/s throughput; Encapsulation: 4,412,249 cycles/op average, 548 ops/s throughput; Decapsulation: 36,017,786 cycles/op average, 67 ops/s throughput. Static/baseline working set memory average: 7,443,486 Bytes; Peak memory average: 7,519,150.08 Bytes

3.3. Transmission and Storage Overhead Testing

Test ID	REF-7
Test Name	CTL-257-512 Reference Implementation Transmission and Storage Overhead Test
Test Objective	Measure the data sizes (public key, private key, ciphertext, shared key) involved in the key encapsulation process for the CTL-257-512 reference implementation.
Prerequisites	1.CTL reference implementation source code runs correctly; 2.Algorithm interfaces can return or confirm the length of various data objects; 3.The test program can output transmission and storage overhead statistics.
Test Steps	1.Navigate to the CTL-257-512 reference implementation source code directory; 2.Run make all to generate benchmark_selfeval_ctl.exe; 3.Execute benchmark_selfeval_ctl.exe; 4.Record public key length, private key length, ciphertext length, and shared key length.
Expected Results	The lengths of all relevant data objects for the CTL-257-512 reference implementation are obtained, in Bytes.
Actual Results	The program runs successfully and outputs transmission and storage overhead test results. The program measures public key length, private key length, ciphertext length, and shared key length over 100 test iterations. Public key length: 521 Bytes; Private key length: 2,953 Bytes; Ciphertext length: 473 Bytes; Shared secret value length: 16 Bytes.

Test ID	REF-8
Test Name	CTL-769-1024 Reference Implementation Transmission and Storage Overhead Test
Test Objective	Measure the data sizes (public key, private key, ciphertext, shared key) involved in the key encapsulation process for the CTL-769-1024 reference implementation.

Prerequisites	1.CTL reference implementation source code runs correctly; 2.Algorithm interfaces can return or confirm the length of various data objects; 3.The test program can output transmission and storage overhead statistics.
Test Steps	1.Navigate to the CTL-769-1024 reference implementation source code directory; 2.Run make all to generate benchmark_selfeval_ctl.exe; 3.Execute benchmark_selfeval_ctl.exe; 4.Record public key length, private key length, ciphertext length, and shared key length.
Expected Results	The lengths of all relevant data objects for the CTL-769-1024 reference implementation are obtained, in Bytes.
Actual Results	The program runs successfully and outputs transmission and storage overhead test results. The program measures public key length, private key length, ciphertext length, and shared key length over 100 test iterations. Public key length: 1,230 Bytes; Private key length: 6,030 Bytes; Ciphertext length: 1,006 Bytes; Shared secret value length: 32 Bytes.

Test ID	REF-9
Test Name	CTL-3329-2048 Reference Implementation Transmission and Storage Overhead Test
Test Objective	Measure the data sizes (public key, private key, ciphertext, shared key) involved in the key encapsulation process for the CTL-3329-2048 reference implementation.
Prerequisites	1.CTL reference implementation source code runs correctly; 2.Algorithm interfaces can return or confirm the length of various data objects; 3.The test program can output transmission and storage overhead statistics.
Test Steps	1.Navigate to the CTL-3329-2048 reference implementation source code directory; 2.Run make all to generate benchmark_selfeval_ctl.exe; 3.Execute benchmark_selfeval_ctl.exe; 4.Record public key length, private key length, ciphertext length, and shared key length.
Expected Results	The lengths of all relevant data objects for the CTL-3329-2048 reference implementation are obtained, in Bytes.
Actual Results	The program runs successfully and outputs transmission and storage overhead test results. The program measures public key length, private key length, ciphertext length, and shared key length over 100 test iterations. Public key length: 3,009 Bytes; Private key length: 15,617 Bytes; Ciphertext length: 2,353 Bytes; Shared secret value length: 48 Bytes.